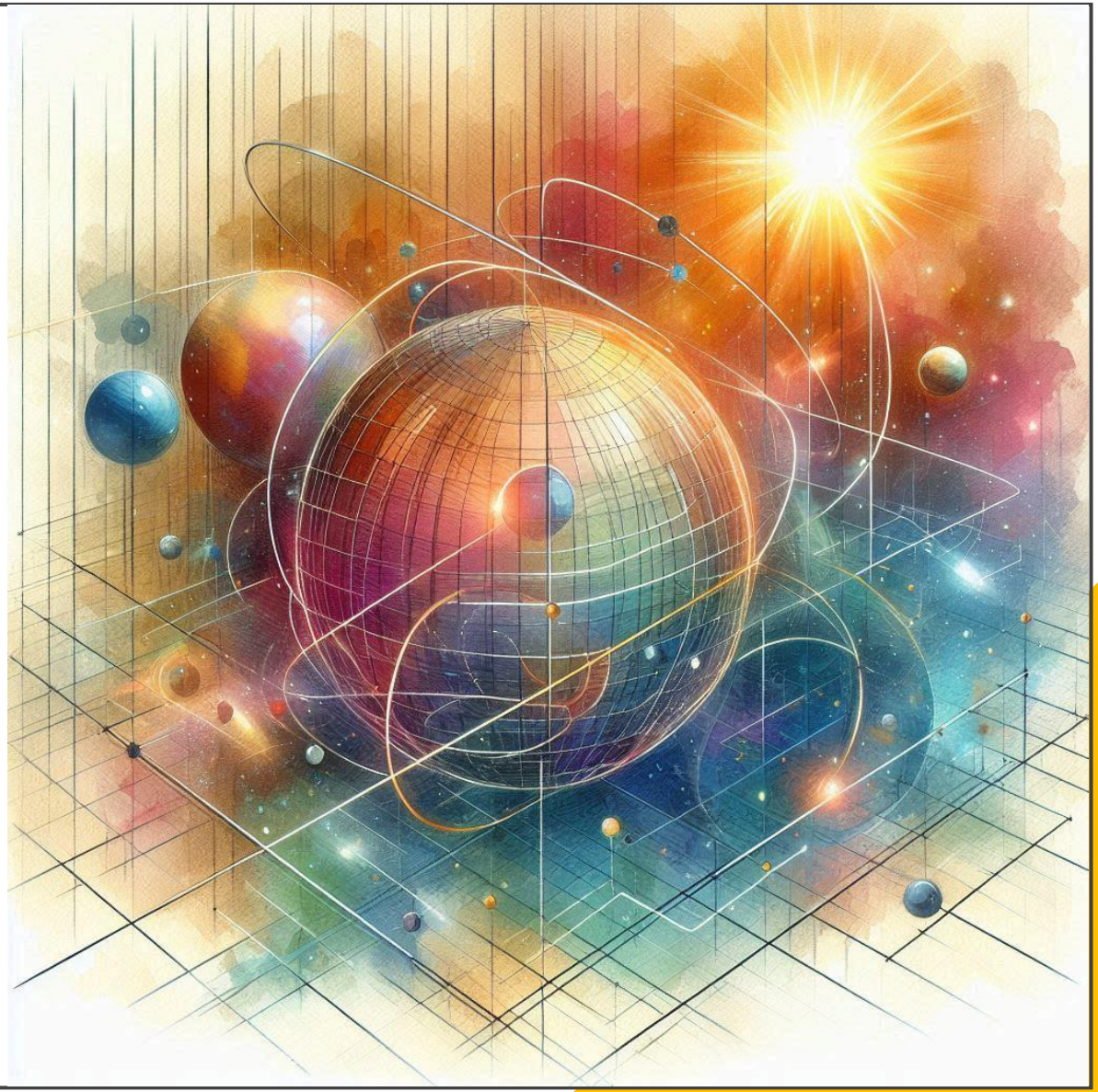
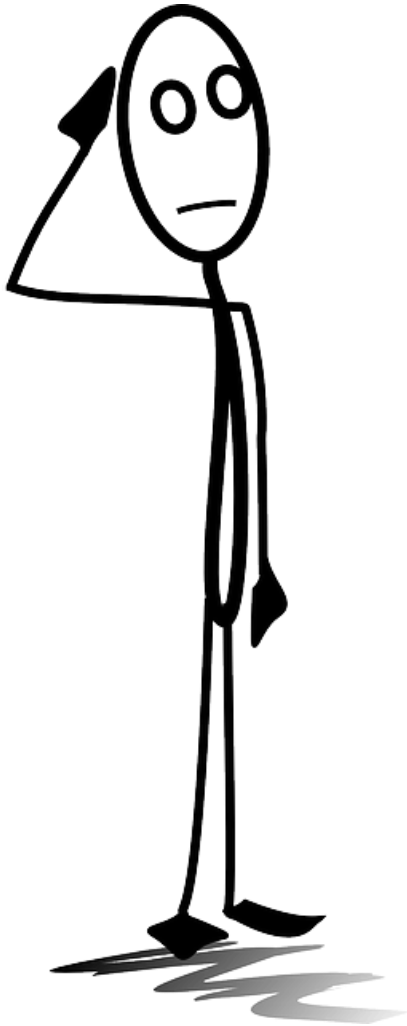
PHYSICS I
WEEK 9



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WEEK 9: OUTLINE

- Review: Curvilinear coordinate systems
- Euler-Lagrange equations (Morin 6.1)



∇ OPERATOR

- ∇ Operator: A vector operator

$$\nabla = \hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}.$$

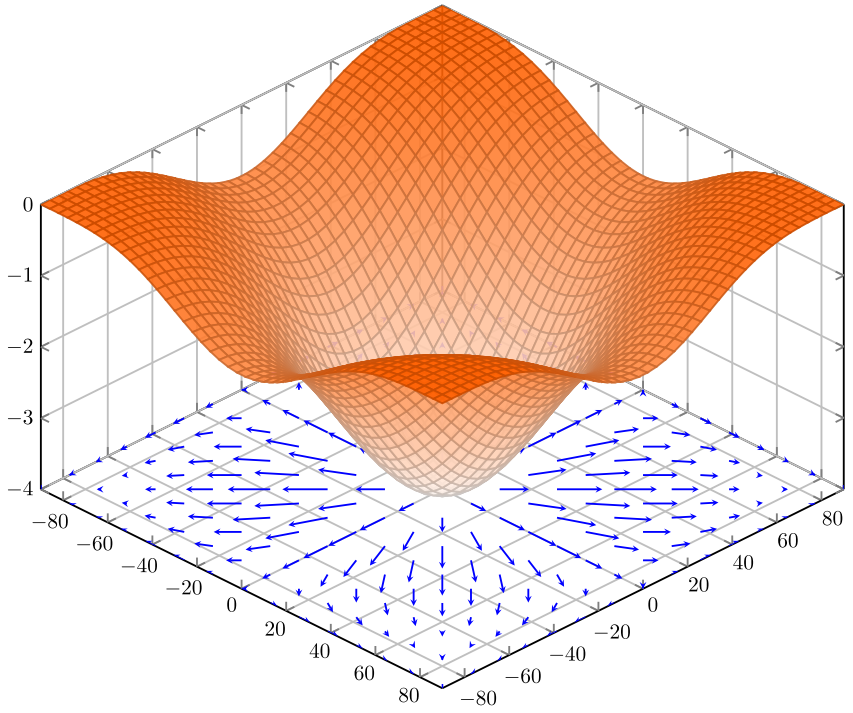
- When acts on a scalar – ∇T: **Gradient**
- When acts on a vector via the dot product – $\nabla . \boldsymbol{T}$: **Divergence**
- When acts on a vector via the cross product– $\nabla \times \boldsymbol{T}$: **Curl**

∇ OPERATOR

- Gradient

$$\nabla T \equiv \frac{\partial T}{\partial x}\hat{\mathbf{x}} + \frac{\partial T}{\partial y}\hat{\mathbf{y}} + \frac{\partial T}{\partial z}\hat{\mathbf{z}}$$

- The magnitude $|\nabla T|$ gives the slope (rate of increase) along the maximal direction
- The gradient ∇T points in the direction of maximum increase of the function T

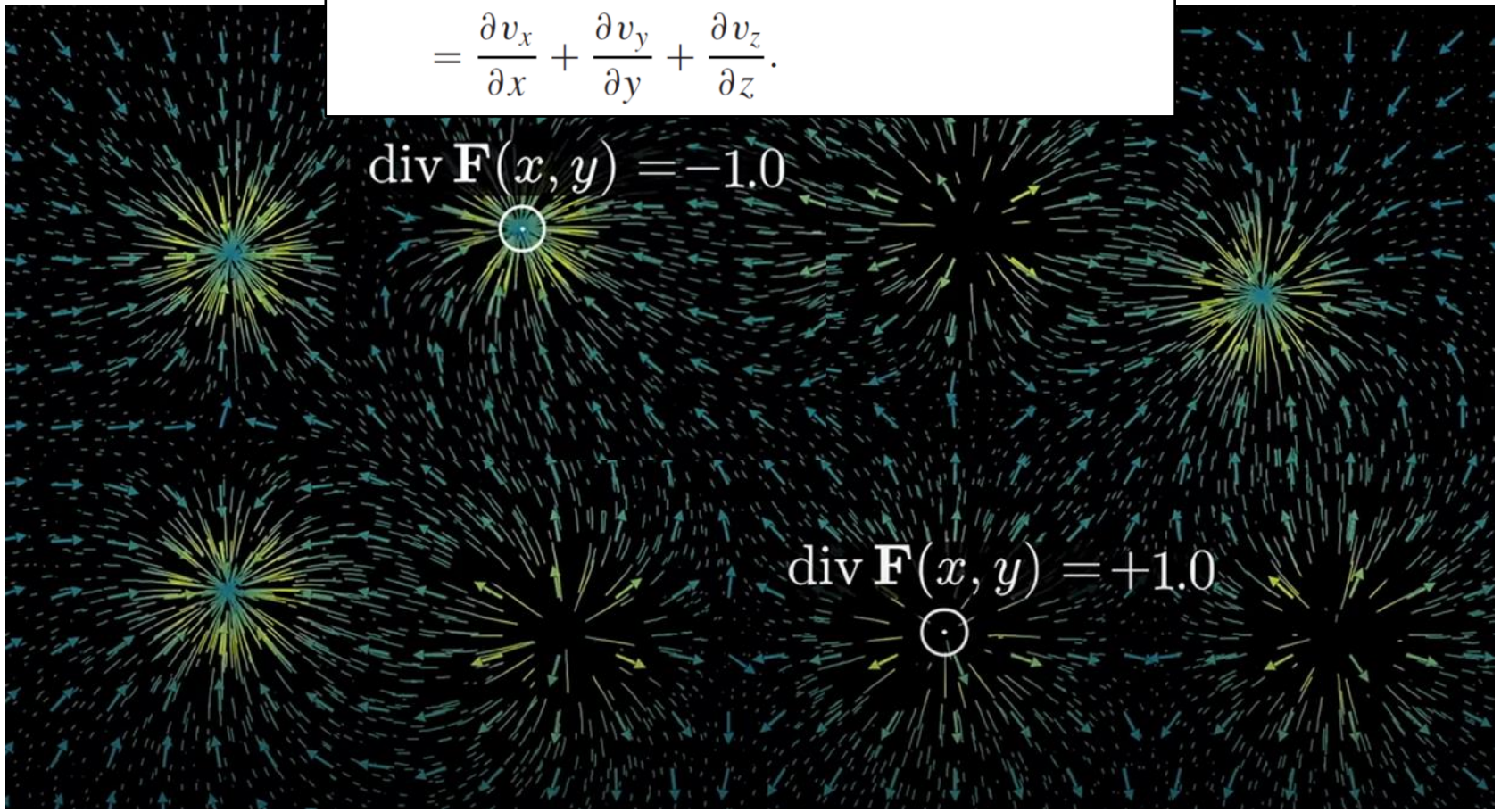


A 2D function

∇ OPERATOR

- Divergence

$$\nabla \cdot \mathbf{v} = \left(\hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}\right) \cdot (v_x\hat{\mathbf{x}} + v_y\hat{\mathbf{y}} + v_z\hat{\mathbf{z}})$$
$$= \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}.$$



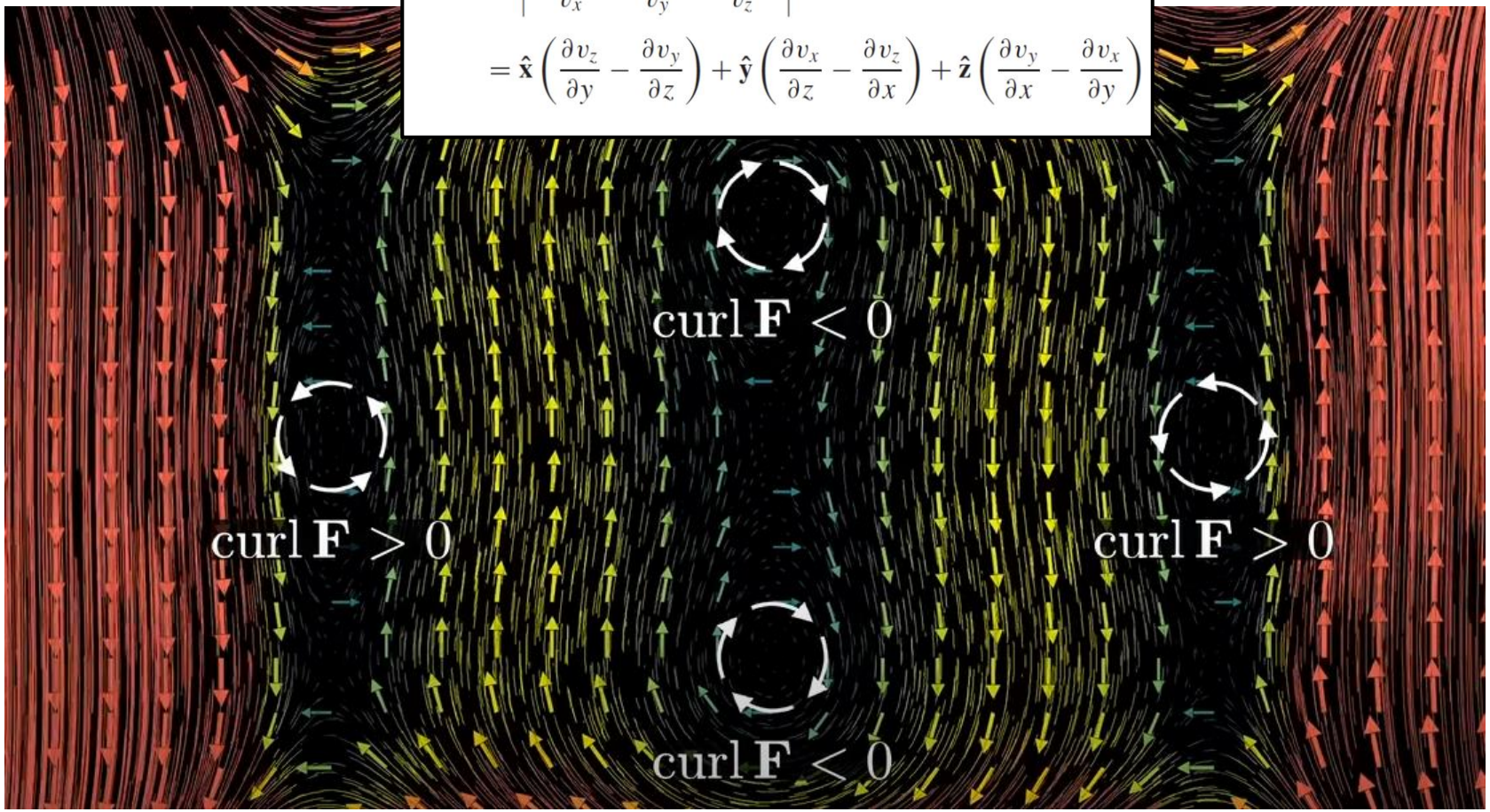
How much the vector spreads out – a scalar quantity

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∇ OPERATOR

- Curl

$$\nabla \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_x & v_y & v_z \end{vmatrix}$$
$$= \hat{\mathbf{x}}\left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)$$



How much the vector curls around – a vector quantity

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CONSERVATIVE FORCE

1) The value of the line integral of a conservative vector field $\boldsymbol{F}(\boldsymbol{r})$ from any point P to any point Q depends only on the end points P and Q and not on the actual path taken from P to Q.

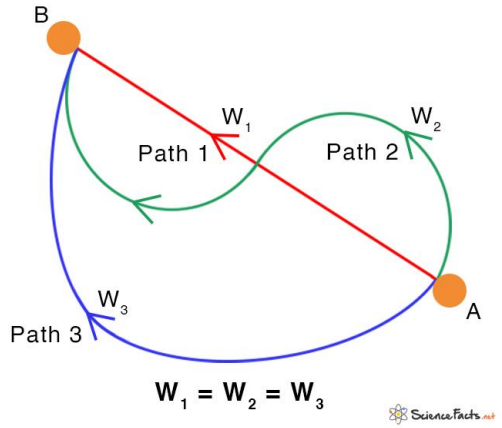
2) The line integral of a conservative vector field $\boldsymbol{F}(\boldsymbol{r})$ around any closed path is equal to zero.

3) A conservative vector field $\boldsymbol{F}(\boldsymbol{r})$ can be written as the gradient of a scalar field,

$$\boldsymbol{F}(\boldsymbol{r}) = -\boldsymbol{\nabla} U(\boldsymbol{r})$$

4) All conservative vector fields are irrotational vector fields,

$$\boldsymbol{\nabla} \times \boldsymbol{F}(\boldsymbol{r}) = -\boldsymbol{\nabla} \times \boldsymbol{\nabla} U(\boldsymbol{r}) = 0$$



SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi \quad , \quad y = r\sin\theta\sin\varphi \quad , \quad z = r\cos\theta$$

Unit vectors:

$$\begin{aligned}
\hat{e}_r &= \sin\theta\cos\varphi\ \hat{e}_x + \sin\theta\sin\varphi\ \hat{e}_y + \cos\theta\ \hat{e}_z \\
\hat{e}_\theta &= \cos\theta\cos\varphi\ \hat{e}_x + \cos\theta\sin\varphi\ \hat{e}_y - \sin\theta\ \hat{e}_z \\
\hat{e}_\varphi &= -\sin\varphi\ \hat{e}_x + \cos\varphi\ \hat{e}_y \quad .
\end{aligned}$$

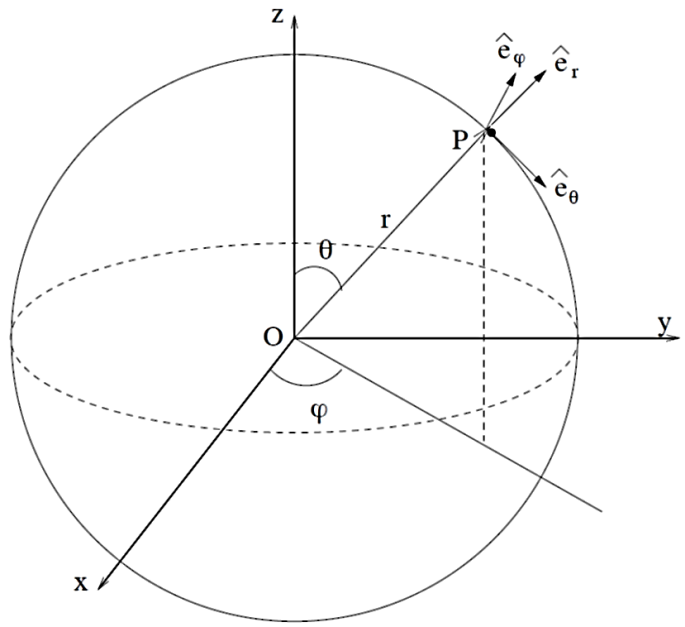
$$\hat{e}_r \times \hat{e}_\theta = \hat{e}_\varphi \ , \ \hat{e}_\theta \times \hat{e}_\varphi = \hat{e}_r \text{ and } \hat{e}_\varphi \times \hat{e}_r = \hat{e}_\theta$$

Position vector: $\vec{r} = r\ \hat{e}_r$

Line element: $\delta\vec{r} = \delta r\ \hat{e}_r + r\ \delta\theta\ \hat{e}_\theta + r\sin\theta\ \delta\varphi\ \hat{e}_\varphi$

Area element: $dS = (R^2\sin\theta\ d\theta\ d\varphi)\ \hat{e}_r$

Volume element: $dV = r^2\sin\theta\ dr\ d\theta\ d\varphi$



The range of values of (r, θ, φ) that cover all points in space are:

$$0 \le r < \infty \quad , \quad 0 \le \theta \le \pi \quad , \quad 0 \le \varphi < 2\pi \ .$$

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi \quad , \quad y = r\sin\theta\sin\varphi \quad , \quad z = r\cos\theta$$

Unit vectors:

$$\begin{aligned}
\hat{e}_r &= \sin\theta\cos\varphi\ \hat{e}_x + \sin\theta\sin\varphi\ \hat{e}_y + \cos\theta\ \hat{e}_z \\
\hat{e}_\theta &= \cos\theta\cos\varphi\ \hat{e}_x + \cos\theta\sin\varphi\ \hat{e}_y - \sin\theta\ \hat{e}_z \\
\hat{e}_\varphi &= -\sin\varphi\ \hat{e}_x + \cos\varphi\ \hat{e}_y \quad .
\end{aligned}$$

$$\hat{e}_r \times \hat{e}_\theta = \hat{e}_\varphi \ , \ \hat{e}_\theta \times \hat{e}_\varphi = \hat{e}_r \text{ and } \hat{e}_\varphi \times \hat{e}_r = \hat{e}_\theta$$

Position vector: $\vec{r} = r\ \hat{e}_r$

Line element: $\delta\vec{r} = \delta r\ \hat{e}_r + r\ \delta\theta\ \hat{e}_\theta + r\sin\theta\ \delta\varphi\ \hat{e}_\varphi$

Area element: $dS = (R^2\sin\theta\ d\theta\ d\varphi)\ \hat{e}_r$

Volume element: $dV = r^2\sin\theta\ dr\ d\theta\ d\varphi$

Total surface area of a sphere with radius R?

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi \quad , \quad y = r\sin\theta\sin\varphi \quad , \quad z = r\cos\theta$$

Unit vectors:

$$\begin{aligned}
\hat{e}_r &= \sin\theta\cos\varphi\ \hat{e}_x + \sin\theta\sin\varphi\ \hat{e}_y + \cos\theta\ \hat{e}_z \\
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\end{aligned}$$

$$\hat{e}_r \times \hat{e}_\theta = \hat{e}_\varphi \ , \ \hat{e}_\theta \times \hat{e}_\varphi = \hat{e}_r \text{ and } \hat{e}_\varphi \times \hat{e}_r = \hat{e}_\theta$$

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Line element: $\delta\vec{r} = \delta r\ \hat{e}_r + r\ \delta\theta\ \hat{e}_\theta + r\sin\theta\ \delta\varphi\ \hat{e}_\varphi$

Area element: $dS = (R^2\sin\theta\ d\theta\ d\varphi)\ \hat{e}_r$

Volume element: $dV = r^2\sin\theta\ dr\ d\theta\ d\varphi$

Total surface area of a sphere with radius R:

$$\int_S dS = \int R^2\ d\Omega = \int_0^\pi d\theta \int_0^{2\pi} d\varphi\ R^2\sin\theta = 4\pi R^2$$

Total angle subtended at the origin: $\int d\Omega = 4\pi$

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi \quad , \quad y = r\sin\theta\sin\varphi \quad , \quad z = r\cos\theta$$

Unit vectors:

$$\begin{aligned}
\hat{e}_r &= \sin\theta\cos\varphi\ \hat{e}_x + \sin\theta\sin\varphi\ \hat{e}_y + \cos\theta\ \hat{e}_z \\
\hat{e}_\theta &= \cos\theta\cos\varphi\ \hat{e}_x + \cos\theta\sin\varphi\ \hat{e}_y - \sin\theta\ \hat{e}_z \\
\hat{e}_\varphi &= -\sin\varphi\ \hat{e}_x + \cos\varphi\ \hat{e}_y \quad .
\end{aligned}$$

$$\hat{e}_r \times \hat{e}_\theta = \hat{e}_\varphi \ , \ \hat{e}_\theta \times \hat{e}_\varphi = \hat{e}_r \text{ and } \hat{e}_\varphi \times \hat{e}_r = \hat{e}_\theta$$

Position vector: $\vec{r} = r\ \hat{e}_r$

Line element: $\delta\vec{r} = \delta r\ \hat{e}_r + r\ \delta\theta\ \hat{e}_\theta + r\sin\theta\ \delta\varphi\ \hat{e}_\varphi$

Area element: $dS = (R^2\sin\theta\ d\theta\ d\varphi)\ \hat{e}_r$

Volume element: $dV = r^2\sin\theta\ dr\ d\theta\ d\varphi$

Total surface area of a sphere with radius R:

$$\int_S dS = \int R^2\ d\Omega = \int_0^\pi d\theta \int_0^{2\pi} d\varphi\ R^2\sin\theta = 4\pi R^2$$

Total angle subtended at the origin: $\int d\Omega = 4\pi$

Total volume of a sphere with radius R?

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi \quad , \quad y = r\sin\theta\sin\varphi \quad , \quad z = r\cos\theta$$

Unit vectors:

$$\begin{aligned}
\hat{e}_r &= \sin\theta\cos\varphi\ \hat{e}_x + \sin\theta\sin\varphi\ \hat{e}_y + \cos\theta\ \hat{e}_z \\
\hat{e}_\theta &= \cos\theta\cos\varphi\ \hat{e}_x + \cos\theta\sin\varphi\ \hat{e}_y - \sin\theta\ \hat{e}_z \\
\hat{e}_\varphi &= -\sin\varphi\ \hat{e}_x + \cos\varphi\ \hat{e}_y \quad .
\end{aligned}$$

$$\hat{e}_r \times \hat{e}_\theta = \hat{e}_\varphi \ , \ \hat{e}_\theta \times \hat{e}_\varphi = \hat{e}_r \text{ and } \hat{e}_\varphi \times \hat{e}_r = \hat{e}_\theta$$

Position vector: $\vec{r} = r\ \hat{e}_r$

Line element: $\delta\vec{r} = \delta r\ \hat{e}_r + r\ \delta\theta\ \hat{e}_\theta + r\sin\theta\ \delta\varphi\ \hat{e}_\varphi$

Area element: $dS = (R^2\sin\theta\ d\theta\ d\varphi)\ \hat{e}_r$

Volume element: $dV = r^2\sin\theta\ dr\ d\theta\ d\varphi$

Total surface area of a sphere with radius R:

$$\int_S dS = \int R^2\ d\Omega = \int_0^\pi d\theta \int_0^{2\pi} d\varphi\ R^2\sin\theta = 4\pi R^2$$

Total angle subtended at the origin: $\int d\Omega = 4\pi$

Total volume of a sphere with radius R:

$$\int_V dV = \int_0^R dr \int_0^\pi d\theta \int_0^{2\pi} d\varphi\ r^2\sin\theta = 4\pi R^3/3$$

SPHERICAL COORDINATE SYSTEM

Position vector $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$$\Rightarrow \boxed{\vec{r} = r\sin\theta\cos\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\theta\,\hat{z}}$$

$$\hat{r} = \frac{\partial\vec{r}/\partial r}{\left|\partial\vec{r}/\partial r\right|} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \frac{\partial\vec{r}/\partial\theta}{\left|\partial\vec{r}/\partial\theta\right|} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\phi} = \frac{\partial\vec{r}/\partial\phi}{\left|\partial\vec{r}/\partial\phi\right|} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$$

$$\frac{d\hat{r}}{dt} = \left(\cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}\right)\dot{\theta} + \sin\theta\left(-\sin\phi\,\hat{x} + \cos\phi\,\hat{y}\right)\dot{\phi}$$

$$= \dot{\theta}\hat{\theta} + \sin\theta\,\dot{\phi}\,\hat{\phi}$$

$$\Rightarrow \boxed{\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\phi}\,\hat{\phi}}$$

Velocity vector

SPHERICAL COORDINATE SYSTEM

Position vector $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$$\Rightarrow \boxed{\vec{r} = r\sin\theta\cos\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\theta\,\hat{z}}$$

$$\hat{r} = \frac{\partial\vec{r}/\partial r}{|\partial\vec{r}/\partial r|} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \frac{\partial\vec{r}/\partial\theta}{|\partial\vec{r}/\partial\theta|} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\phi} = \frac{\partial\vec{r}/\partial\phi}{|\partial\vec{r}/\partial\phi|} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$$

$$\frac{d\hat{r}}{dt} = \left(\cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}\right)\dot{\theta} + \sin\theta\left(-\sin\phi\,\hat{x} + \cos\phi\,\hat{y}\right)\dot{\phi}$$

$$= \dot{\theta}\hat{\theta} + \sin\theta\,\dot{\phi}\,\hat{\phi}$$

$$\Rightarrow \boxed{\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\phi}\,\hat{\phi}}$$

Velocity vector

Kinetic Energy:

$$T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2)$$

Acceleration:

$$\boldsymbol{a} = \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \frac{\mathrm{d}\left(\dot{r}\,\hat{\boldsymbol{r}}\right)}{\mathrm{d}t} + \frac{\mathrm{d}\left(r\,\dot{\theta}\,\hat{\boldsymbol{\theta}}\right)}{\mathrm{d}t} + \frac{\mathrm{d}\left(r\sin\theta\,\dot{\phi}\,\hat{\boldsymbol{\phi}}\right)}{\mathrm{d}t}$$

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi \quad , \quad y = r\sin\theta\sin\varphi \quad , \quad z = r\cos\theta$$

Unit vectors:

$$\begin{aligned}
\hat{e}_r &= \sin\theta\cos\varphi\ \hat{e}_x + \sin\theta\sin\varphi\ \hat{e}_y + \cos\theta\ \hat{e}_z \\
\hat{e}_\theta &= \cos\theta\cos\varphi\ \hat{e}_x + \cos\theta\sin\varphi\ \hat{e}_y - \sin\theta\ \hat{e}_z \\
\hat{e}_\varphi &= -\sin\varphi\ \hat{e}_x + \cos\varphi\ \hat{e}_y \quad .
\end{aligned}$$

$$\hat{e}_r \times \hat{e}_\theta = \hat{e}_\varphi \ , \ \hat{e}_\theta \times \hat{e}_\varphi = \hat{e}_r \text{ and } \hat{e}_\varphi \times \hat{e}_r = \hat{e}_\theta$$

Position vector: $\vec{r} = r\ \hat{e}_r$

Line element: $\delta\vec{r} = \delta r\ \hat{e}_r + r\ \delta\theta\ \hat{e}_\theta + r\sin\theta\ \delta\varphi\ \hat{e}_\varphi$

Area element: $dS = (R^2\sin\theta\ d\theta\ d\varphi)\ \hat{e}_r$

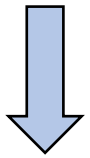
Volume element: $dV = r^2\sin\theta\ dr\ d\theta\ d\varphi$

Gradient: $\vec{\nabla} = \hat{e}_r\frac{\partial}{\partial r} + \hat{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta} + \hat{e}_\varphi\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}$

Kinetic Energy:

$$T = \tfrac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2)$$

$$\boldsymbol{\nabla} U(x,y,z) = \frac{\partial U}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial U}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial U}{\partial z}\hat{\boldsymbol{z}}$$



$$\boldsymbol{\nabla} U(r,\theta,\emptyset) = \frac{\partial U}{\partial r}\hat{\boldsymbol{r}} + \frac{1}{r}\frac{\partial U}{\partial\theta}\widehat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\frac{\partial U}{\partial\emptyset}\widehat{\emptyset}$$



SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi \quad , \quad y = r\sin\theta\sin\varphi \quad , \quad z = r\cos\theta$$

Unit vectors:

$$\begin{aligned}
\hat{e}_r &= \sin\theta\cos\varphi\ \hat{e}_x + \sin\theta\sin\varphi\ \hat{e}_y + \cos\theta\ \hat{e}_z \\
\hat{e}_\theta &= \cos\theta\cos\varphi\ \hat{e}_x + \cos\theta\sin\varphi\ \hat{e}_y - \sin\theta\ \hat{e}_z \\
\hat{e}_\varphi &= -\sin\varphi\ \hat{e}_x + \cos\varphi\ \hat{e}_y \quad .
\end{aligned}$$

$$\hat{e}_r \times \hat{e}_\theta = \hat{e}_\varphi \ , \ \hat{e}_\theta \times \hat{e}_\varphi = \hat{e}_r \text{ and } \hat{e}_\varphi \times \hat{e}_r = \hat{e}_\theta$$

Position vector: $\vec{r} = r\ \hat{e}_r$

Line element: $\delta\vec{r} = \delta r\ \hat{e}_r + r\ \delta\theta\ \hat{e}_\theta + r\sin\theta\ \delta\varphi\ \hat{e}_\varphi$

Area element: $dS = (R^2\sin\theta\ d\theta\ d\varphi)\ \hat{e}_r$

Volume element: $dV = r^2\sin\theta\ dr\ d\theta\ d\varphi$

Gradient: $\vec{\nabla} = \hat{e}_r\frac{\partial}{\partial r} + \hat{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta} + \hat{e}_\varphi\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}$

Kinetic Energy:

$$T = \tfrac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2)$$

Conservative central force:

$$\mathbf{F(r)} = -\nabla U = -\hat{\mathbf{r}}\frac{\partial U}{\partial r} - \hat{\boldsymbol{\theta}}\frac{1}{r}\frac{\partial U}{\partial\theta} - \hat{\boldsymbol{\phi}}\frac{1}{r\sin\theta}\frac{\partial U}{\partial\phi}$$

⇩

$$\mathbf{F(r)} = -\hat{\mathbf{r}}\frac{\partial U}{\partial r}$$

Only the radial component should be non-zero

⇩

$$\partial U/\partial\theta = \partial U/\partial\phi = 0$$

Spherically symmetric!

The magnitude only depends on the separation

CYLINDRICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

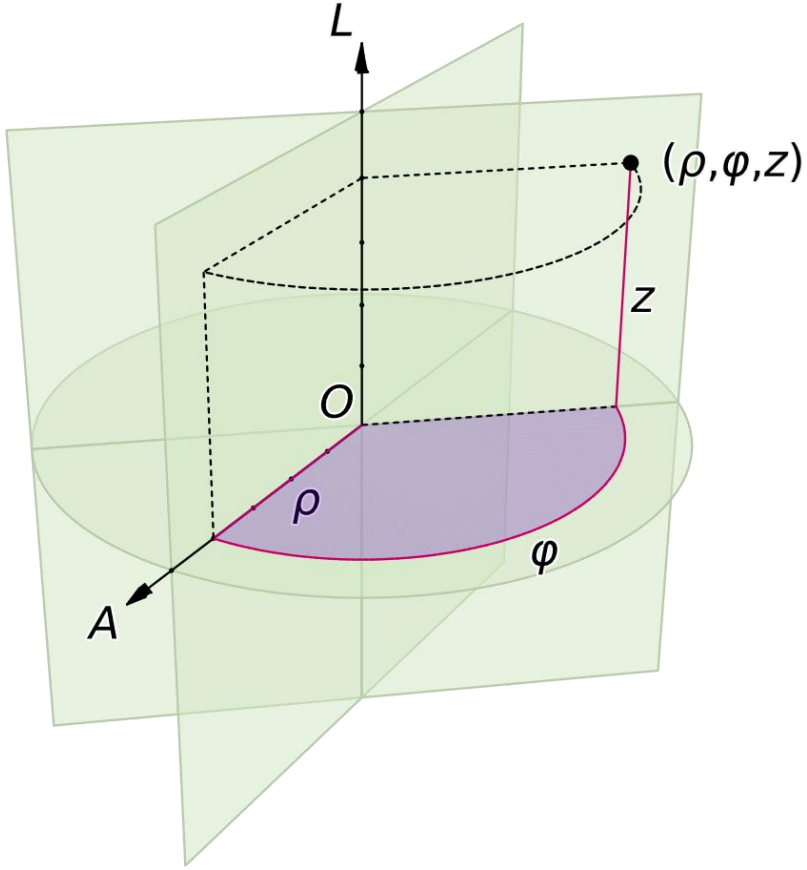
$$x = \rho\cos\varphi \qquad y = \rho\sin\varphi \qquad z = z$$

Unit vectors:

$$\begin{aligned}
\hat{e}_\rho &= \cos\varphi\ \hat{e}_x + \sin\varphi\ \hat{e}_y \\
\hat{e}_\varphi &= -\sin\varphi\ \hat{e}_x + \cos\varphi\ \hat{e}_y \\
\hat{e}_z &= \hat{e}_z
\end{aligned}$$

$$\hat{e}_\rho \times \hat{e}_\varphi = \hat{e}_z\ , \quad \hat{e}_\varphi \times \hat{e}_z = \hat{e}_\rho\ , \quad \hat{e}_z \times \hat{e}_\rho = \hat{e}_\varphi$$

$$\begin{aligned}
\rho &= +\sqrt{x^2 + y^2} \\
\varphi &= \tan^{-1}(y/x) \\
z &= z
\end{aligned}$$

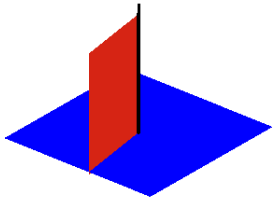


Position vector: $\vec{r} = \rho\ \hat{e}_\rho + z\ \hat{e}_z$

Line element: $\delta\vec{r} = \delta\rho\ \hat{e}_\rho + \rho\ \delta\varphi\ \hat{e}_\varphi + \delta z\ \hat{e}_z$

Area element: $\rho\ d\rho\ d\varphi\ \hat{e}_z$, $\rho\ d\varphi\ dz\ \hat{e}_\rho$ and $d\rho\ dz\ \hat{e}_\phi$

Volume element: $dV = (d\rho)\ (\rho d\varphi)\ (dz) = \rho\ d\rho\ d\varphi\ dz$



CYLINDRICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = \rho\cos\varphi \qquad y = \rho\sin\varphi \qquad z = z$$

Unit vectors:

$$\begin{aligned}\hat{e}_\rho &= \cos\varphi\ \hat{e}_x + \sin\varphi\ \hat{e}_y\\ \hat{e}_\varphi &= -\sin\varphi\ \hat{e}_x + \cos\varphi\ \hat{e}_y\\ \hat{e}_z &= \hat{e}_z\end{aligned}$$

Relations with the spherical coordinates:

$$\begin{aligned}\hat{e}_r &= \sin\theta\ \hat{e}_\rho + \cos\theta\ \hat{e}_z\\ \hat{e}_\theta &= \cos\theta\ \hat{e}_\rho - \sin\theta\ \hat{e}_z\\ \hat{e}_\varphi &= \hat{e}_\varphi\ .\end{aligned}$$

$$\hat{e}_\rho \times \hat{e}_\varphi = \hat{e}_z\ , \quad \hat{e}_\varphi \times \hat{e}_z = \hat{e}_\rho\ , \quad \hat{e}_z \times \hat{e}_\rho = \hat{e}_\varphi$$

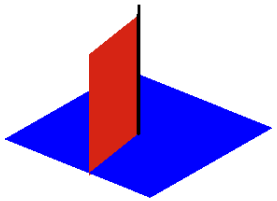
Position vector: $\vec{r} = \rho\ \hat{e}_\rho + z\ \hat{e}_z$

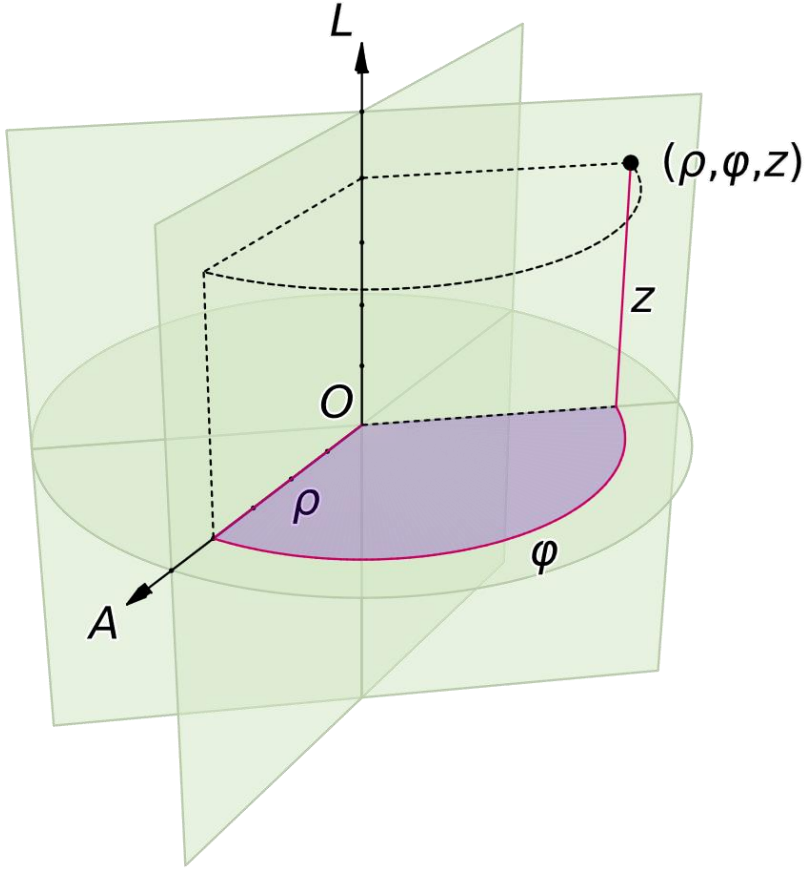
Line element: $\delta\vec{r} = \delta\rho\ \hat{e}_\rho + \rho\ \delta\varphi\ \hat{e}_\varphi + \delta z\ \hat{e}_z$

Area element: $\rho\ d\rho\ d\varphi\ \hat{e}_z,\ \rho\ d\varphi\ dz\ \hat{e}_\rho$ and $d\rho\ dz\ \hat{e}_\phi$

Volume element: $dV = (d\rho)\ (\rho d\varphi)\ (dz) = \rho\ d\rho\ d\varphi\ dz$

Gradient: $\vec{\nabla} = \hat{e}_\rho\ \frac{\partial}{\partial\rho} + \hat{e}_\varphi\ \frac{1}{\rho}\frac{\partial}{\partial\varphi} + \hat{e}_z\ \frac{\partial}{\partial z}$





CYLINDRICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = \rho\cos\varphi \qquad y = \rho\sin\varphi \qquad z = z$$

Unit vectors:

$$\begin{aligned}\hat{e}_\rho &= \cos\varphi\ \hat{e}_x + \sin\varphi\ \hat{e}_y\\ \hat{e}_\varphi &= -\sin\varphi\ \hat{e}_x + \cos\varphi\ \hat{e}_y\\ \hat{e_z} &= \hat{e_z}\end{aligned}$$

$$\hat{e}_\rho \times \hat{e}_\varphi = \hat{e}_z\ , \quad \hat{e}_\varphi \times \hat{e}_z = \hat{e}_\rho\ , \quad \hat{e}_z \times \hat{e}_\rho = \hat{e}_\varphi$$

Relations with the spherical coordinates:

$$\begin{aligned}\hat{e}_r &= \sin\theta\ \hat{e}_\rho + \cos\theta\ \hat{e}_z\\ \hat{e}_\theta &= \cos\theta\ \hat{e}_\rho - \sin\theta\ \hat{e}_z\\ \hat{e}_\varphi &= \hat{e}_\varphi\ .\end{aligned}$$

Position vector: $\vec{r} = \rho\ \hat{e}_\rho + z\ \hat{e}_z$

Line element: $\delta\vec{r} = \delta\rho\ \hat{e}_\rho + \rho\ \delta\varphi\ \hat{e}_\varphi + \delta z\ \hat{e}_z$

Area element: $\rho\, d\rho\, d\varphi\ \hat{e}_z,\ \rho\, d\varphi\, dz\ \hat{e}_\rho$ and $d\rho\, dz\ \hat{e}_\phi$

Volume element: $dV = (d\rho)\ (\rho d\varphi)\ (dz) = \rho\ d\rho\ d\varphi\ dz$

Gradient: $\vec{\nabla} = \hat{e}_\rho\ \dfrac{\partial}{\partial\rho} + \hat{e}_\varphi\ \dfrac{1}{\rho}\dfrac{\partial}{\partial\varphi} + \hat{e}_z\ \dfrac{\partial}{\partial z}$

Kinetic Energy:

$$T = \frac{m}{2}[(\dot{\rho})^2 + (\rho\dot{\phi})^2 + (\dot{z})^2]$$

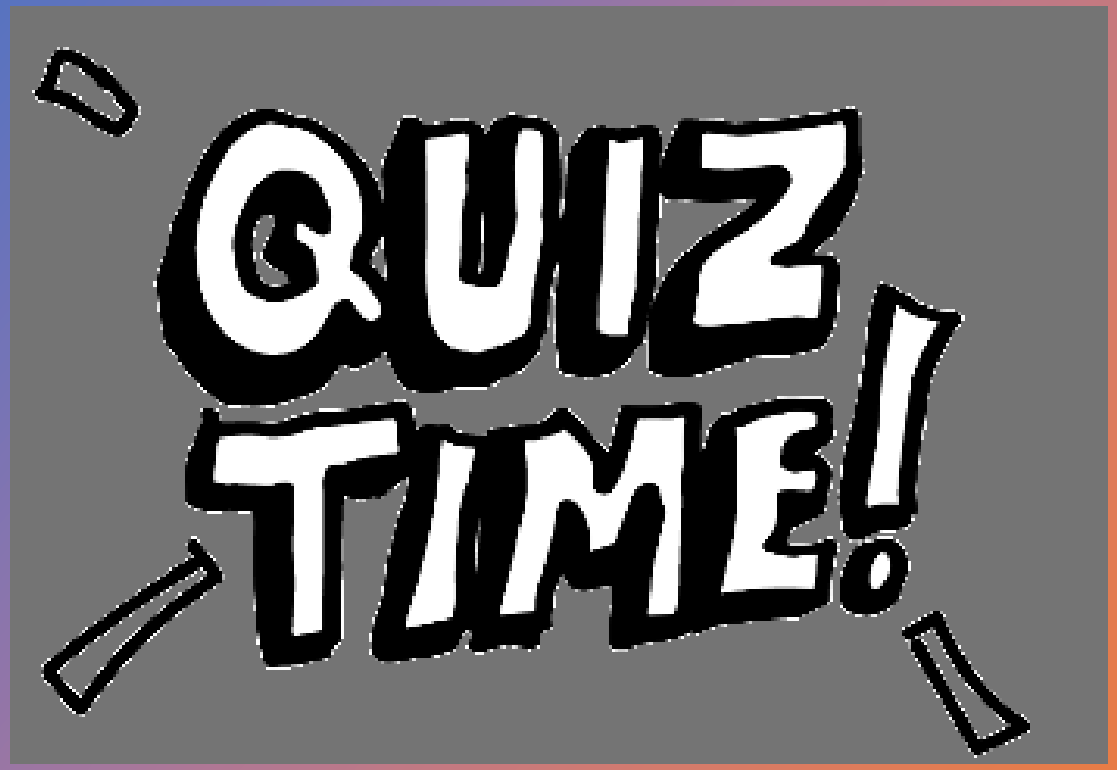
Total surface area of a cylinder with radius R and height L :

$$\int_0^L dz \int_0^{2\pi} R\, d\varphi + 2\int_0^R \rho\, d\rho \int_0^{2\pi} d\varphi = 2\pi RL + 2\pi R^2$$

Total volume of a cylinder with radius R and height L:

$$\int_0^L dz \int_0^R \rho\ d\rho \int_0^{2\pi} d\varphi = \pi R^2 L$$

https://en.wikipedia.org/wiki/Del_in_cylindrical_and_spherical_coordinates



QUIZ 2

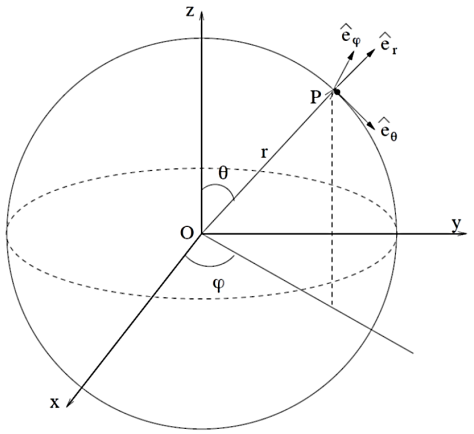
10TH OCTOBER 2024

- **Unit 3** – Motion in one dimension and Phase Space
- **Unit 4** – Work-energy theorem in 3-dimensions, Conservative forces
- Material covered in Tutorials 4, 5, 6
- Expect short questions on **Unit 5**

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$



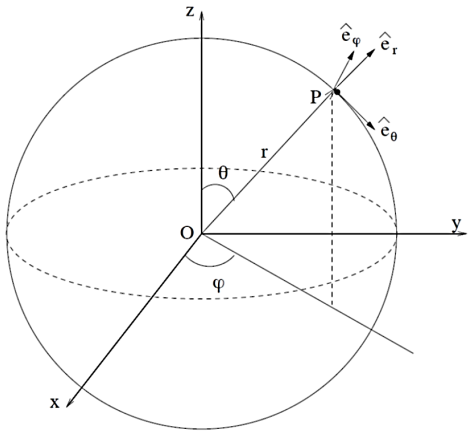
Unit vectors:

$$\hat{\mathbf{r}} = \sin\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) + \cos\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\theta}} = \cos\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) - \sin\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\varphi}} = -\sin\varphi\hat{\mathbf{x}} + \cos\varphi\hat{\mathbf{y}}$$

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$



Unit vectors:

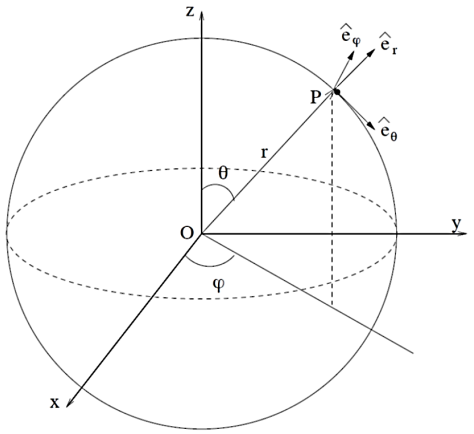
$$\hat{\mathbf{r}} = \sin\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) + \cos\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\theta}} = \cos\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) - \sin\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\varphi}} = -\sin\varphi\hat{\mathbf{x}} + \cos\varphi\hat{\mathbf{y}}$$

$$\hat{\mathbf{x}} = \sin\theta\cos\varphi\hat{\mathbf{r}} + \cos\theta\cos\varphi\hat{\boldsymbol{\theta}} - \sin\varphi\hat{\boldsymbol{\varphi}}$$
$$\hat{\mathbf{y}} = \sin\theta\sin\varphi\hat{\mathbf{r}} + \cos\theta\sin\varphi\hat{\boldsymbol{\theta}} + \cos\varphi\hat{\boldsymbol{\varphi}}$$
$$\hat{\mathbf{z}} = \cos\theta\hat{\mathbf{r}} - \sin\theta\hat{\boldsymbol{\theta}}$$

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$



Unit vectors:

$$\hat{\mathbf{r}} = \sin\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) + \cos\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\theta}} = \cos\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) - \sin\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\varphi}} = -\sin\varphi\hat{\mathbf{x}} + \cos\varphi\hat{\mathbf{y}}$$

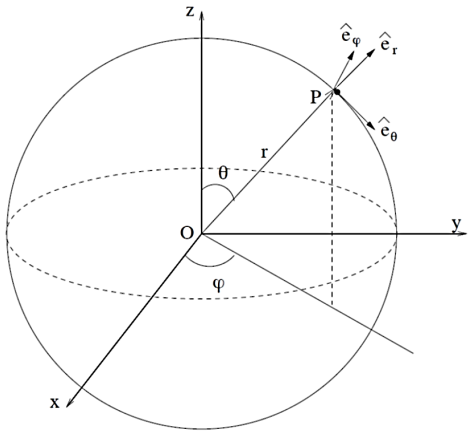
$$\vec{\nabla} = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}$$

$$\hat{\mathbf{x}} = \sin\theta\cos\varphi\hat{\mathbf{r}} + \cos\theta\cos\varphi\hat{\boldsymbol{\theta}} - \sin\varphi\hat{\boldsymbol{\varphi}}$$
$$\hat{\mathbf{y}} = \sin\theta\sin\varphi\hat{\mathbf{r}} + \cos\theta\sin\varphi\hat{\boldsymbol{\theta}} + \cos\varphi\hat{\boldsymbol{\varphi}}$$
$$\hat{\mathbf{z}} = \cos\theta\hat{\mathbf{r}} - \sin\theta\hat{\boldsymbol{\theta}}$$

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$



Unit vectors:

$$\hat{\mathbf{r}} = \sin\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) + \cos\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\theta}} = \cos\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) - \sin\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\varphi}} = -\sin\varphi\hat{\mathbf{x}} + \cos\varphi\hat{\mathbf{y}}$$

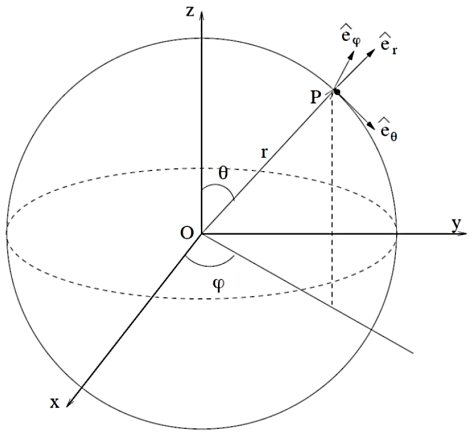
$$\vec{\nabla} = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} = a\hat{r}\frac{\partial}{\partial r} + b\hat{\theta}\frac{\partial}{\partial\theta} + c\hat{\phi}\frac{\partial}{\partial\phi}$$

$$\hat{\mathbf{x}} = \sin\theta\cos\varphi\hat{\mathbf{r}} + \cos\theta\cos\varphi\hat{\boldsymbol{\theta}} - \sin\varphi\hat{\boldsymbol{\varphi}}$$
$$\hat{\mathbf{y}} = \sin\theta\sin\varphi\hat{\mathbf{r}} + \cos\theta\sin\varphi\hat{\boldsymbol{\theta}} + \cos\varphi\hat{\boldsymbol{\varphi}}$$
$$\hat{\mathbf{z}} = \cos\theta\hat{\mathbf{r}} - \sin\theta\hat{\boldsymbol{\theta}}$$

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$



Unit vectors:

$$\hat{\mathbf{r}} = \sin\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) + \cos\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\theta}} = \cos\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) - \sin\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\varphi}} = -\sin\varphi\hat{\mathbf{x}} + \cos\varphi\hat{\mathbf{y}}$$

$$\hat{\mathbf{x}} = \sin\theta\cos\varphi\hat{\mathbf{r}} + \cos\theta\cos\varphi\hat{\boldsymbol{\theta}} - \sin\varphi\hat{\boldsymbol{\varphi}}$$
$$\hat{\mathbf{y}} = \sin\theta\sin\varphi\hat{\mathbf{r}} + \cos\theta\sin\varphi\hat{\boldsymbol{\theta}} + \cos\varphi\hat{\boldsymbol{\varphi}}$$
$$\hat{\mathbf{z}} = \cos\theta\hat{\mathbf{r}} - \sin\theta\hat{\boldsymbol{\theta}}$$

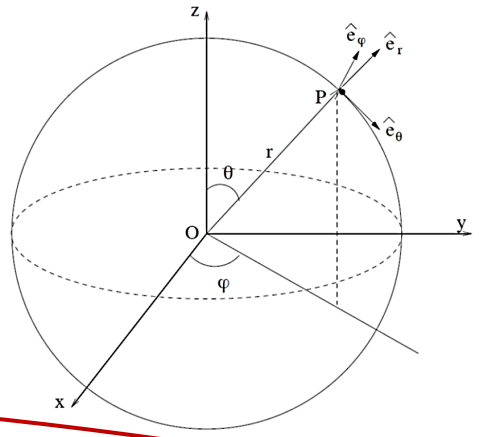
$$\vec{\nabla} = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} = a\hat{r}\frac{\partial}{\partial r} + b\hat{\theta}\frac{\partial}{\partial\theta} + c\hat{\phi}\frac{\partial}{\partial\phi}$$

$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r}\frac{\partial}{\partial x} + \frac{\partial y}{\partial r}\frac{\partial}{\partial y} + \frac{\partial z}{\partial r}\frac{\partial}{\partial z}$$
$$\frac{\partial}{\partial\theta} = \frac{\partial x}{\partial\theta}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\theta}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\theta}\frac{\partial}{\partial z}$$
$$\frac{\partial}{\partial\phi} = \frac{\partial x}{\partial\phi}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\phi}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\phi}\frac{\partial}{\partial z}$$

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$



Unit vectors:

$$\hat{\mathbf{r}} = \sin\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) + \cos\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\theta}} = \cos\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) - \sin\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\varphi}} = -\sin\varphi\hat{\mathbf{x}} + \cos\varphi\hat{\mathbf{y}}$$

$$\hat{\mathbf{x}} = \sin\theta\cos\varphi\hat{\mathbf{r}} + \cos\theta\cos\varphi\hat{\boldsymbol{\theta}} - \sin\varphi\hat{\boldsymbol{\varphi}}$$
$$\hat{\mathbf{y}} = \sin\theta\sin\varphi\hat{\mathbf{r}} + \cos\theta\sin\varphi\hat{\boldsymbol{\theta}} + \cos\varphi\hat{\boldsymbol{\varphi}}$$
$$\hat{\mathbf{z}} = \cos\theta\hat{\mathbf{r}} - \sin\theta\hat{\boldsymbol{\theta}}$$

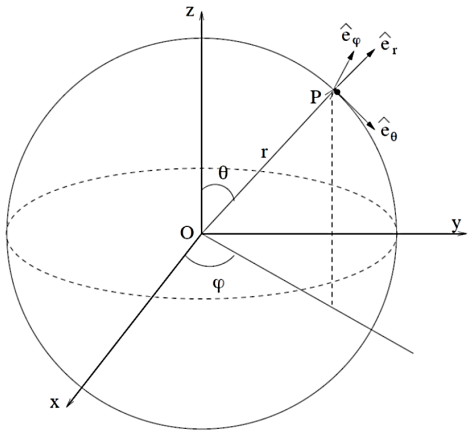
$$\vec{\nabla} = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} = a\hat{r}\frac{\partial}{\partial r} + b\hat{\theta}\frac{\partial}{\partial\theta} + c\hat{\phi}\frac{\partial}{\partial\phi}$$

$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r}\frac{\partial}{\partial x} + \frac{\partial y}{\partial r}\frac{\partial}{\partial y} + \frac{\partial z}{\partial r}\frac{\partial}{\partial z}$$
$$\frac{\partial}{\partial\theta} = \frac{\partial x}{\partial\theta}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\theta}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\theta}\frac{\partial}{\partial z}$$
$$\frac{\partial}{\partial\phi} = \frac{\partial x}{\partial\phi}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\phi}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\phi}\frac{\partial}{\partial z}$$

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$



Unit vectors:

$$\hat{\mathbf{r}} = \sin\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) + \cos\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\theta}} = \cos\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) - \sin\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\varphi}} = -\sin\varphi\hat{\mathbf{x}} + \cos\varphi\hat{\mathbf{y}}$$

$$\hat{\mathbf{x}} = \sin\theta\cos\varphi\hat{\mathbf{r}} + \cos\theta\cos\varphi\hat{\boldsymbol{\theta}} - \sin\varphi\hat{\boldsymbol{\varphi}}$$
$$\hat{\mathbf{y}} = \sin\theta\sin\varphi\hat{\mathbf{r}} + \cos\theta\sin\varphi\hat{\boldsymbol{\theta}} + \cos\varphi\hat{\boldsymbol{\varphi}}$$
$$\hat{\mathbf{z}} = \cos\theta\hat{\mathbf{r}} - \sin\theta\hat{\boldsymbol{\theta}}$$

$$\vec{\nabla} = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z} = a\hat{r}\frac{\partial}{\partial r} + b\hat{\theta}\frac{\partial}{\partial\theta} + c\hat{\phi}\frac{\partial}{\partial\phi}$$

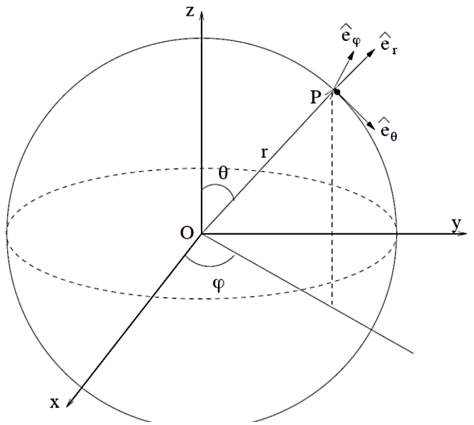
$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r}\frac{\partial}{\partial x} + \frac{\partial y}{\partial r}\frac{\partial}{\partial y} + \frac{\partial z}{\partial r}\frac{\partial}{\partial z}$$
$$\frac{\partial}{\partial\theta} = \frac{\partial x}{\partial\theta}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\theta}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\theta}\frac{\partial}{\partial z}$$
$$\frac{\partial}{\partial\phi} = \frac{\partial x}{\partial\phi}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\phi}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\phi}\frac{\partial}{\partial z}$$

$$\boldsymbol{\nabla} f = \frac{\partial f}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial f}{\partial\theta}\hat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\frac{\partial f}{\partial\varphi}\hat{\boldsymbol{\varphi}}$$

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$



Unit vectors:

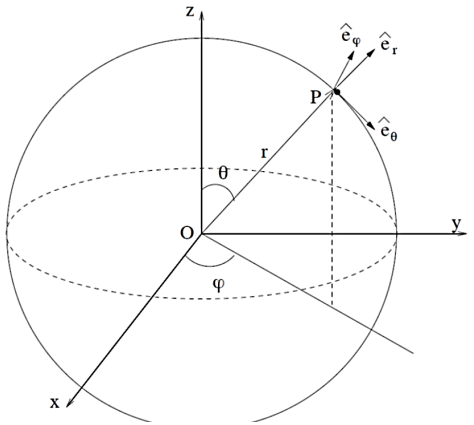
$$\hat{\mathbf{r}} = \sin\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) + \cos\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\theta}} = \cos\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) - \sin\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\varphi}} = -\sin\varphi\hat{\mathbf{x}} + \cos\varphi\hat{\mathbf{y}}$$

$$\dot{\hat{\mathbf{r}}} = (\cos\theta\dot{\theta}\cos\phi - \sin\theta\sin\phi\dot{\phi})\hat{\mathbf{x}} + (\cos\theta\dot{\theta}\sin\phi + \sin\theta\cos\phi\dot{\phi})\hat{\mathbf{y}} - \sin\theta\dot{\theta}\hat{\mathbf{z}}$$

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$



Unit vectors:

$$\hat{\mathbf{r}} = \sin\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) + \cos\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\theta}} = \cos\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) - \sin\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\varphi}} = -\sin\varphi\hat{\mathbf{x}} + \cos\varphi\hat{\mathbf{y}}$$

$$\dot{\hat{\mathbf{r}}} = (\cos\theta\dot{\theta}\cos\phi - \sin\theta\sin\phi\dot{\phi})\hat{\mathbf{x}} + (\cos\theta\dot{\theta}\sin\phi + \sin\theta\cos\phi\dot{\phi})\hat{\mathbf{y}} - \sin\theta\dot{\theta}\hat{\mathbf{z}}$$

$$\dot{\hat{\mathbf{r}}} = \dot{\theta}\hat{\boldsymbol{\theta}} + \sin\theta\dot{\phi}\hat{\boldsymbol{\phi}}$$

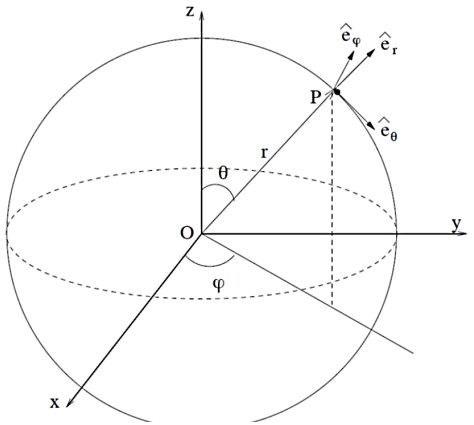
$$\dot{\hat{\boldsymbol{\theta}}} = \cos\theta\dot{\phi}\hat{\boldsymbol{\phi}} - \dot{\theta}\hat{\mathbf{r}}$$

$$\dot{\hat{\boldsymbol{\phi}}} = -\sin\theta\dot{\phi}\hat{\mathbf{r}} - \cos\theta\dot{\phi}\hat{\boldsymbol{\theta}}$$

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$



Unit vectors:

$$\hat{\mathbf{r}} = \sin\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) + \cos\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\theta}} = \cos\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) - \sin\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\varphi}} = -\sin\varphi\hat{\mathbf{x}} + \cos\varphi\hat{\mathbf{y}}$$

$$\dot{\hat{\mathbf{r}}} = (\cos\theta\dot{\theta}\cos\phi - \sin\theta\sin\phi\dot{\phi})\hat{\mathbf{x}} + (\cos\theta\dot{\theta}\sin\phi + \sin\theta\cos\phi\dot{\phi})\hat{\mathbf{y}} - \sin\theta\dot{\theta}\hat{\mathbf{z}}$$

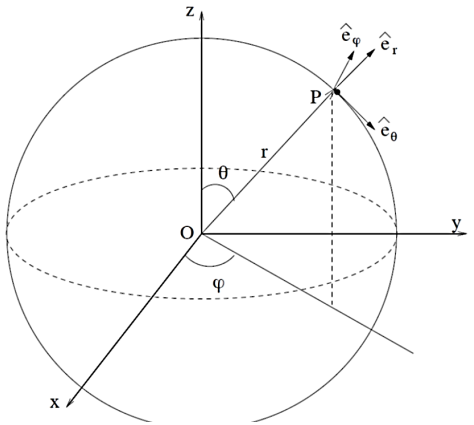
$$\dot{\hat{\mathbf{r}}} = \dot{\theta}\hat{\boldsymbol{\theta}} + \sin\theta\dot{\phi}\hat{\boldsymbol{\phi}} \qquad \dot{\hat{\boldsymbol{\theta}}} = \cos\theta\dot{\phi}\hat{\boldsymbol{\phi}} - \dot{\theta}\hat{\mathbf{r}} \qquad \dot{\hat{\boldsymbol{\phi}}} = -\sin\theta\dot{\phi}\hat{\mathbf{r}} - \cos\theta\dot{\phi}\hat{\boldsymbol{\theta}}$$

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r}\hat{\mathbf{r}} + r\dot{\hat{\mathbf{r}}} = \dot{r}\hat{r} + r(\dot{\theta}\hat{\boldsymbol{\theta}} + \sin\theta\dot{\phi}\hat{\boldsymbol{\phi}})$$
$$= \dot{r}\hat{\mathbf{r}} + r\,\dot{\theta}\hat{\boldsymbol{\theta}} + r\sin\theta\dot{\phi}\hat{\boldsymbol{\phi}}$$

SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$



Unit vectors:

$$\hat{\mathbf{r}} = \sin\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) + \cos\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\theta}} = \cos\theta\left(\cos\varphi\hat{\mathbf{x}} + \sin\varphi\hat{\mathbf{y}}\right) - \sin\theta\hat{\mathbf{z}}$$
$$\hat{\boldsymbol{\varphi}} = -\sin\varphi\hat{\mathbf{x}} + \cos\varphi\hat{\mathbf{y}}$$

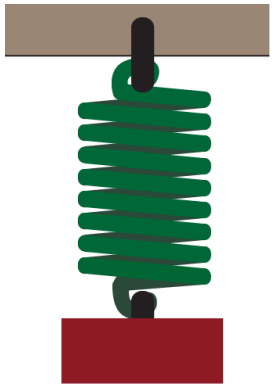
$$\dot{\hat{\mathbf{r}}} = (\cos\theta\dot{\theta}\cos\phi - \sin\theta\sin\phi\dot{\phi})\hat{\mathbf{x}} + (\cos\theta\dot{\theta}\sin\phi + \sin\theta\cos\phi\dot{\phi})\hat{\mathbf{y}} - \sin\theta\dot{\theta}\hat{\mathbf{z}}$$

$$\dot{\hat{\mathbf{r}}} = \dot{\theta}\hat{\boldsymbol{\theta}} + \sin\theta\dot{\phi}\hat{\boldsymbol{\phi}} \qquad \dot{\hat{\boldsymbol{\theta}}} = \cos\theta\dot{\phi}\hat{\boldsymbol{\phi}} - \dot{\theta}\hat{\mathbf{r}} \qquad \dot{\hat{\boldsymbol{\phi}}} = -\sin\theta\dot{\phi}\hat{\mathbf{r}} - \cos\theta\dot{\phi}\hat{\boldsymbol{\theta}}$$

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r}\hat{\mathbf{r}} + r\dot{\hat{\mathbf{r}}} = \dot{r}\hat{r} + r(\dot{\theta}\hat{\boldsymbol{\theta}} + \sin\theta\dot{\phi}\hat{\boldsymbol{\phi}})$$
$$= \dot{r}\hat{\mathbf{r}} + r\,\dot{\theta}\hat{\boldsymbol{\theta}} + r\sin\theta\dot{\phi}\hat{\boldsymbol{\phi}}$$

$$\mathbf{a} = \dot{\mathbf{v}} = \ddot{r}\hat{\mathbf{r}} + \dot{r}(\dot{\theta}\hat{\boldsymbol{\theta}} + \sin\theta\dot{\phi}\hat{\boldsymbol{\phi}}) + \dot{r}\dot{\theta}\hat{\boldsymbol{\theta}} + r\ddot{\theta}\hat{\boldsymbol{\theta}} + r\dot{\theta}(\cos\theta\dot{\phi}\hat{\boldsymbol{\phi}} - \dot{\theta}\hat{\mathbf{r}}) + \dot{r}\sin\theta\dot{\phi}\hat{\boldsymbol{\phi}} + r\cos\theta\dot{\theta}\dot{\phi}\hat{\boldsymbol{\phi}} + r\sin\theta\ddot{\phi}\hat{\boldsymbol{\phi}} + r\sin$$
$$\theta\dot{\phi}(-\sin\theta\dot{\phi}\hat{\mathbf{r}} - \cos\theta\dot{\phi}\hat{\boldsymbol{\theta}})$$

EULER-LAGRANGE EQUATIONS

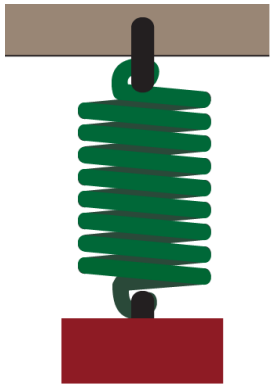


Newton' law

$$F = ma$$

$$m\ddot{x} = -kx$$

EULER-LAGRANGE EQUATIONS



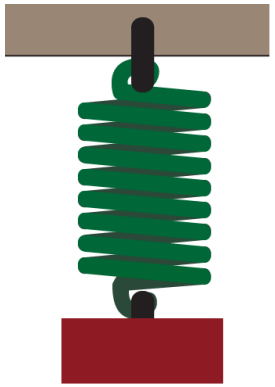
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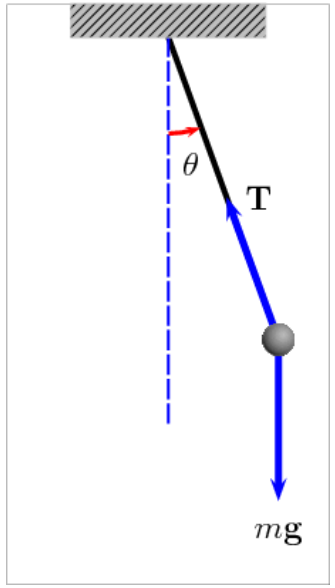
$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\partial L}{\partial x}$$

Euler-Lagrange equations

EULER-LAGRANGE EQUATIONS

Simple pendulum

$$x = l\sin\theta$$
$$y = l\cos\theta$$



$$T = \frac{1}{2}mv^2 \qquad U = mgl(1-\cos\theta)$$

$$L = T - U$$

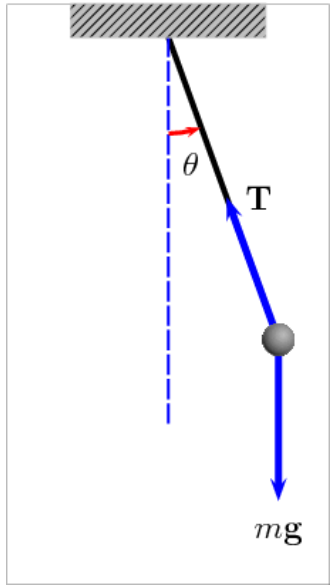
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Euler-Lagrange equations

$$\ddot{\theta} = -\frac{g}{l}\sin\theta$$